FINM3013 Introduction to Financial Derivatives

Chapter 4 Exercises

Solutions

1. The forward rate is

$$\frac{0.06\times 0.75 - 0.05\times 0.50}{0.25} = 0.08$$

or 8%. The FRA rate is 7%. A profit can therefore be made by borrowing for six months at 5%, entering into an FRA to borrow for the period between 6 and 9 months for 7% and lending for nine months at 6%. The profit is $e^{0.06(0.75)} - e^{0.05(0.5)}e^{0.07(0.25)} = 0.002612$ for every dollar borrowed.

2.

a) With continuous compounding the 6-month rate is $2\ln 1.02 = 0.039605$ or 3.961%. The 12-month rate is $2\ln 1.0225 = 0.044501$ or 4.4501%. The 18-month rate is $2\ln 1.02375 = 0.046945$ or 4.6945%. The 24-month rate is $2\ln 1.025 = 0.049385$ or 4.9385%.

b) The forward rate (expressed with continuous compounding) is

$$\frac{4.9385\times 2 - 4.6945\times 1.5}{0.5}$$

or 5.6707%. When expressed with semiannual compounding this is $2(e^{0.056707\times 0.5} - 1) = 0.057518$ or 5.7518%.

c) The value of an FRA that promises to pay 6% for the six month period starting in 18 months is

$$1{,}000{,}000\times(0.06 - 0.057518)\times 0.5e^{-0.049385\times 2} = 1{,}124$$

or \$1,124.

3.

a) The zero rate for a maturity of six months, expressed with continuous compounding is $2\ln(1+2/98) = 4.0405\%$. The zero rate for a maturity of one year, expressed with continuous compounding is $\ln(1+5/95) = 5.1293$. The 1.5-year rate is R where

$$3.1e^{-0.040405\times 0.5} + 3.1e^{-0.051293\times 1} + 103.1e^{-R\times 1.5} = 101$$

The solution to this equation is $R = 0.054429$. The 2.0-year rate is R where

$$4e^{-0.040405\times 0.5} + 4e^{-0.051293\times 1} + 4e^{-0.054429\times 1.5} + 104e^{-R\times 2} = 104$$

The solution to this equation is $R = 0.058085$. These results are shown in the table below

Maturity (yrs)	*Zero Rate (%)*	*Forward Rate (%)*	*Par Yield (s.a.%)*	*Par yield (c.c %)*
0.5	4.0405	4.0405	4.0816	4.0405
1.0	5.1293	6.2181	5.1813	5.1154
1.5	5.4429	6.0700	5.4986	5.4244
2.0	5.8085	6.9054	5.8620	5.7778

b) The continuously compounded forward rates are shown in the third column of the table

c) The par yield, expressed with semiannual compounding, is shown in the fourth column of the table. In the fifth column of the table it is converted to continuous compounding

d) The price of the bond is

$$3.5e^{-0.040405\times 0.5}+3.5e^{-0.051293\times 1}+3.5e^{-0.054429\times 1.5}+103.5e^{-0.058085\times 2}=102.13$$

The yield on the bond, y satisfies

$$3.5e^{-y\times 0.5}+3.5e^{-y\times 1.0}+3.5e^{-y\times 1.5}+103.5e^{-y\times 2.0}=102.13$$

The solution to this equation is $y=0.057723$. The bond yield is therefore 5.7723%.